Consistent hypothesis and version space.

An hypothesis h is Consistent with a set of training examples D iff $h(x) = c(x)$ for each example in D

$$\text{Consistent}(h, D) \equiv (\forall \langle x, c(x) \rangle \in D) \; h(x) = c(x))$$

Eg

	Citations	Size	In Library	price	Editions	By
1	some	small	NO	affordable	one	NO
2	many	Big	NO	Expensive	Many	Yes

any variable name.

$h_1 = (?, ?, NO, ?, many)$ — Consistent

$h_2 = (?, ?, NO, ?, ?)$ — Inconsistent

The version space $VS_{H,D}$ is the subset of the hypothesis from H consistent with the training example in D (training examples)

$$VS_{H,D} = \{ h \in H \mid \text{Consistent}(h, D) \}$$

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List-Then-Eliminate Algorithm

Version space as list of hypotheses

1) Version space - a list containing every hypothesis in H.

2) For each training example, $\langle x, c(x) \rangle$

Remove from Versionspace any hypothesis h for which $h(x) \neq c(x)$

3) Output the list of hypotheses in version space.

Eg: features (attributes)

$F_1 \rightarrow A, B$ (possibilities)

$F_2 \rightarrow X, Y$ (?, $\emptyset$) Consider.

Instance space : (A, X) (A, Y), (B, Y),

(B, X) (B, $\emptyset$) (B, ?)

Hypothesis space : (A, X) (A, Y) (A, $\emptyset$)

(A, ?), (B, X) (B, Y), (B, $\emptyset$)

(B, ?), ($\emptyset$, X), ($\emptyset$, Y), ($\emptyset$, $\emptyset$), ($\emptyset$, ?)

(?, X), (?, Y), (?, $\emptyset$), (?, ?) = 16 Hypothesis

Semantically distinct Hypothesis :-

(A, X), (A, Y), (A, ?)(B, X), (B, Y)

(B, ?) (?, X), (?, Y), (?, ?),

(∅, ∅) = 10

Version space :- list All Hypothesis in given data

= (A, X)(A, Y), (A, ?), (B, X)

(B, Y), (B, ?)(?, X), (?, Y), (?, ?)

(∅, ∅).

Eg	F_1	F_2	Target
	A	X	yes
	A	Y	yes

Case : (A, X) → 1st +ve

(A, Y) → X not matched Expected +ve

not consistent

(A, X) removed.

(A, ?) — which consistent with both examples with return.

(?, ?) — two consistent Hypothesis

Problems

(i) The hypothesis space must be finite

(ii) Enumeration of all the hypothesis, Rather inefficient.

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